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COURSE CODE: SBT-DF204-BASIC NETWORKING SKILLS

LAB NAME: Case Study 1 Investigating Harassment Email Traffic with Wireshark

DATE: 05/10/26

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Computer Forensics Case Studies

Case Study 1 Investigating Harassment Email Traffic With Wireshark Student Assessment Brief

Assessment detail	Information
Course	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 1 Individual forensic investigation
Case title	Investigating Harassment Email Traffic With Wireshark
Submission format	Written report with supporting screenshots and evidence log
Deadline and maximum score	As published in E-Campus

This individual case study asks you to investigate a simulated harassment-email complaint using supplied email material, a class roster, and a packet capture. Your report must show how you preserved the evidence, found the relevant web activity in Wireshark, and assessed whether the traffic supports attribution to a student in Chemistry 109. Do not assume that an IP address proves identity. Your conclusion must follow the evidence and explain its limits.

Case scenario

Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use. On Monday 21 July, a later message arrived through the

web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109.

Case materials

Use the scenario-provided first email, full headers, second email, class roster, and network traffic capture.

Scenario page: digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario

Traffic capture: digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap

If the source page also provides a published checksum, compare it with the hash you calculate. If it does not, record your calculated SHA-256 value in the report so another examiner can identify the evidence file you analysed.

What you must submit

Item	Required output	What to provide
1	Forensic working copy	Download the scenario traffic capture and work only from your copied evidence file. Record the original file name, source link, file size, download date, and SHA-256 value.
2	Wireshark analysis	Use display filters, packet inspection, and Follow TCP Stream where appropriate. Preserve packet numbers and exact timestamps for every material finding.
3	Evidence-based conclusion	State only what the traffic supports. Separate direct observations, reasonable inferences, and limitations caused by the shared open wireless network.
4	Student report	Submit one report in PDF or DOCX format, with an evidence log and clearly labelled screenshots in an appendix or separate ZIP file.

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kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/harass
Session Actions Edit View Help
Monday 05 October 2026

(root@kali)~[/harass]
cd
ls
forensic_case_Lab1_SBT-DF_203 forensic_case_SBT-DF203 SBT-DF203-Lab3 SBT-DF203-Lab7 secret.txt
forensic_case_Lab1_SBT-DF203 harass SBT-DF203-Lab4 SBT-DF203-Lab8 stegano_lab4
forensic_case_SBT-DF_203 SBT-DF203-Lab2 SBT-DF203-Lab5 SBT-DF203-Lab9 _tower_original_image_for_lab.bmp

(root@kali)~[/harass]
cd harass
ls
(root@kali)~[/harass]
wget -q https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap

Prepended http:// to '-q'
--2026-10-05 13:32:46-- http://xn--q-5gn/
Resolving xn--q-5gn (xn--q-5gn)... failed: Name or service not known.
wget: unable to resolve host address 'xn--q-5gn'
--2026-10-05 13:33:11-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 16.12.102.1, 16.15.42.12, 52.92.160.225, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|16.12.102.1|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap'

nitroba.pcap          100%[=====] 53.58M  2.38MB/s  in 27s

2026-10-05 13:33:40 (1.98 MB/s) - 'nitroba.pcap' saved [56180821/56180821]

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

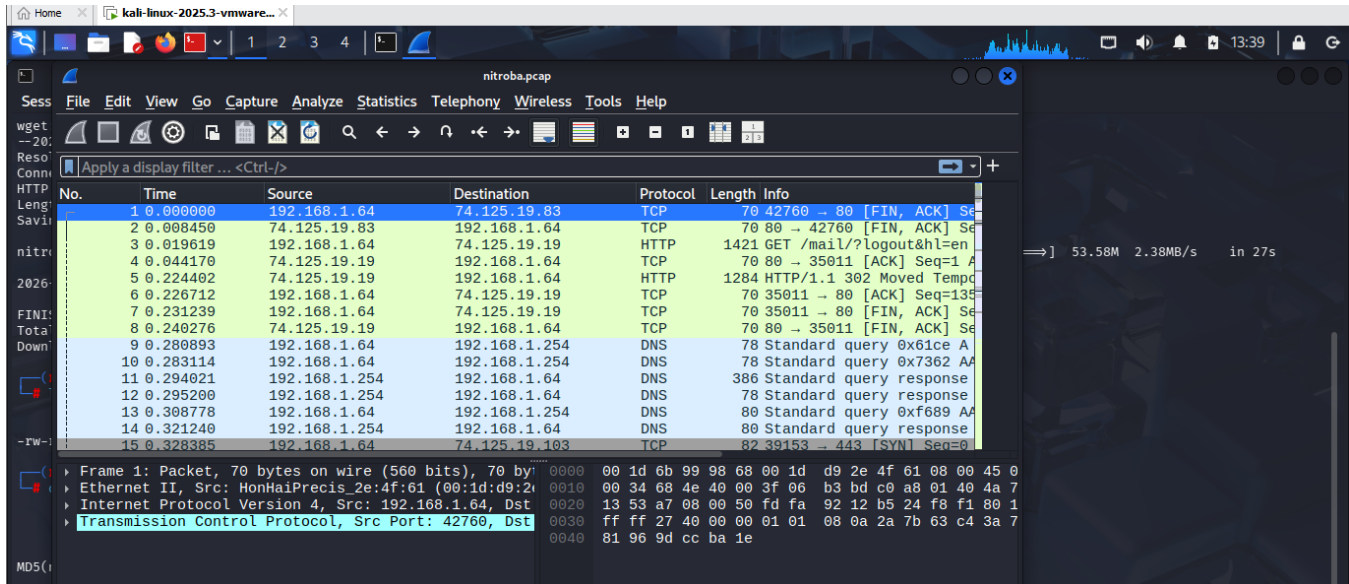
FINISHED --2026-10-05 13:33:40--
Total wall clock time: 54s
Downloaded: 1 files, 54M in 27s (1.98 MB/s)

(root@kali) ~/harass
ls -l nitroba.pcap
-rw-rw-r-- 1 root root 56180821 Nov 21 2020 nitroba.pcap

(root@kali) ~/harass
openssl dgst -md5 nitroba.pcap
MD5(nitroba.pcap)= 9981827f11968773ff815e39f5458ec8

(root@kali) ~/harass
wireshark nitroba.pcap
Warning: program compiled against libxml 215 using older 214

```



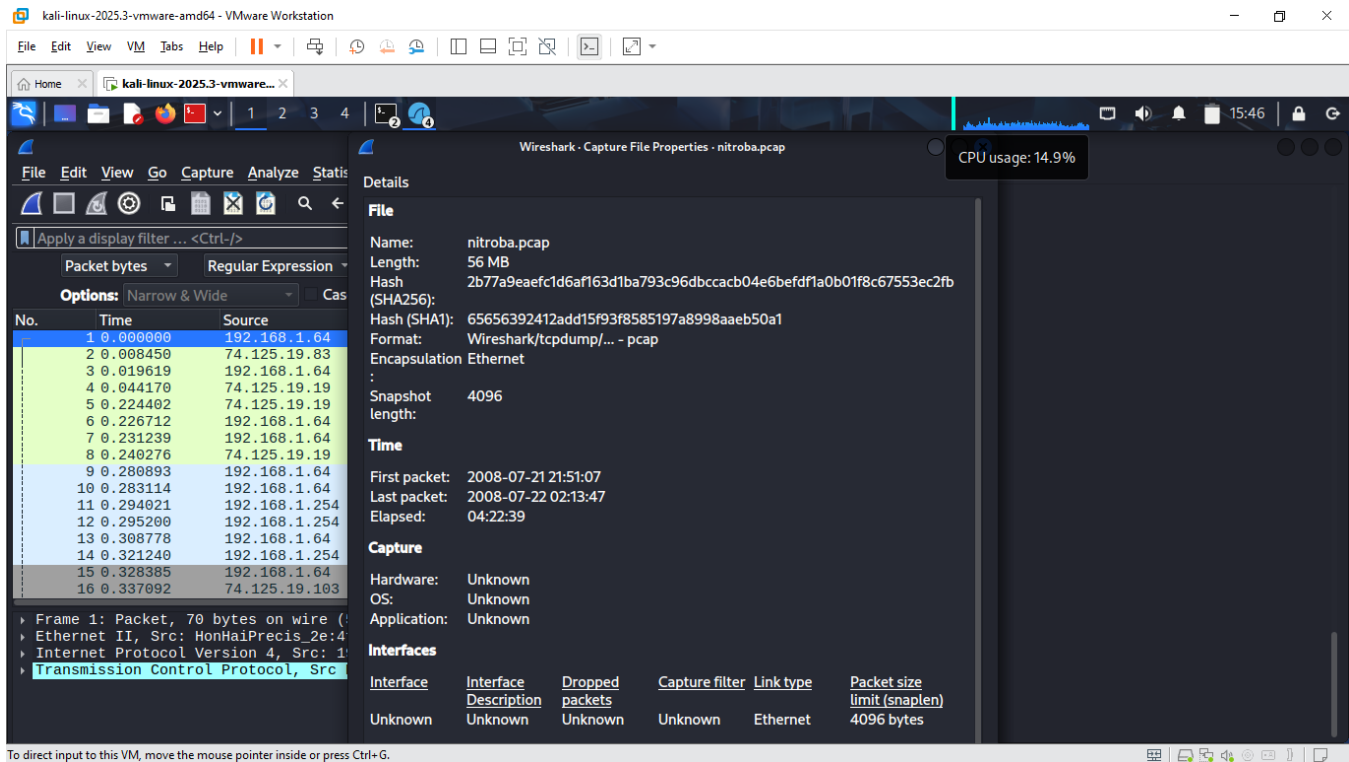
Investigation task

Conduct a methodical forensic review of the packet capture. Use the questions below as the structure for your findings. For every material finding, provide the Wireshark display filter, packet number or stream reference, timestamp, and a short explanation of why the evidence matters. You may use the filters as starting points, but test and adapt them against the actual capture.

No	Investigation question	Minimum evidence to discuss
1	What file did you acquire, and how did you preserve its integrity?	Give the file name: nitroba.pcap File source: Nitroba University File Size: 54M in 27s (1.98 MB/s) File SHA-256 value: 2b77a9eaefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb Method you used to create a working copy: ROOT@KALI

No	Investigation question	Minimum evidence to discuss
2	Which client system contacted the web service?	Identify the Client IP: 192.168.1.64 Service IP: 74.125.19.84 Packet number: 94410 Filter: ip:192.168.1.64, service ip:74.125.19.84, port: 80
3	What evidence links the client to the harassment message?	Show the relevant HTTP POST and explain how its content, form field, URI, or stream correlates with the reported message.
4	Which device made the relevant request?	Identify the MAC address associated with the client traffic and explain why a shared IP address alone cannot prove identity. The MAC address is 00:1d:d9:2e:4f:61 Why the IP address alone cannot prove identity: ANSWER: The ip address 192.168.1.64 identify a network address but not a particular person. In a shared environment an ip address can be used by multiple over time and NAT/DHCP can also cause address to be shared.
5	Who is associated with the device?	Use only scenario-provided cookie or browser evidence to link the device to an identity. State the packet evidence and avoid unsupported assumptions. ANSWER: Packet 1 identifies the client device/network interface as MAC 00:1d:d9:2e:4f:61, associated with IP 192.168.1.64. However, the available packet evidence does not establish the identity of a particular person. There is no provided cookie or browser evidence linking this device to an individual is visible, so no personal identity can be derive.
6	Is that person on the Chem 109 roster?	Compare the supported identity with the provided roster and show the basis for your answer. ANSWER: The packet supports that the traffic came from the device with MAC Address 00:1d:d9:2e:4f:61, using IP 192.168.1.64. It does not support identifying the person behind that device. Therefore, a roster entry should only be matched if the roster explicitly maps this MAC address
7	When did the activity occur?	Create a short timeline using exact PCAP timestamps. State the time zone displayed by Wireshark and any conversion you make. ANSWER:

No	Investigation question	Minimum evidence to discuss
8	What conclusion can you defend?	Give a careful conclusion, confidence level, and at least two limitations or alternative explanations. ANSWER: The PCAP provides strong evidence that the observed traffic originated from the client device using IP 192.168.1.64 and MAC address 00:1d:d9:2e:4f:61. Packet 1 shows the client communicating with the service IP 74.125.19.83. In as much as, the evidence does not establish the identity of a particular person. No scenario-provided cookie, browser credential, or other unique user identifier has been provided that links the device to a named individual in the roster. Therefore, it would be inappropriate to attribute the activity to a specific person based only on the IP or MAC address. Confidence level: • High confidence that 192.168.1.64 / 00:1d:d9:2e:4f:61 represents the observed client ip. Limitations and Explanations ✓ Shared or reassigned IP address: 192.168.1.64 may have been assigned through DHCP or shared by different users at different times. An IP address alone does not identify a person. ✓ MAC address identifies an interface, not a Human: The MAC address identifies the network interface observed in the capture, but another person could have been using the same computer. ✓ MISSING IDENTITY EVIDENCE: Without a scenario-provided cookie, browser session identifier, login information, or explicit roster-to-MAC mapping, there is no reliable basis for naming a person. ✓ LIMITATIONS: The visible timestamps are relative PCAP times rather than an absolute time-of-day, so they cannot independently establish who was using the device at a particular real-world time.



Wireshark starting points

The following filters support the workflow shown in the case material. Replace the values in angle brackets with values you derive from the capture. Do not treat a filter result as a conclusion until you inspect the packet details and document the connection to the investigation question.

Purpose	Example display filter
Locate web requests	<code>http.request.method == "GET" && http.host == "www.willselfdestruct.com"</code>
Search page content	<code>data-text-lines contains "secure anonymous E-mail"</code>
Identify form activity	<code>http.request.method == "POST"</code>
Narrow to the relevant exchange	<code>ip.src == <client_ip> && ip.dst == <service_ip> && http.request.method == "POST"</code>
Trace a device	<code>eth.addr == <client_mac></code>
Look for identity evidence	<code>eth.src == <client_mac> && http.cookie contains "@"</code>

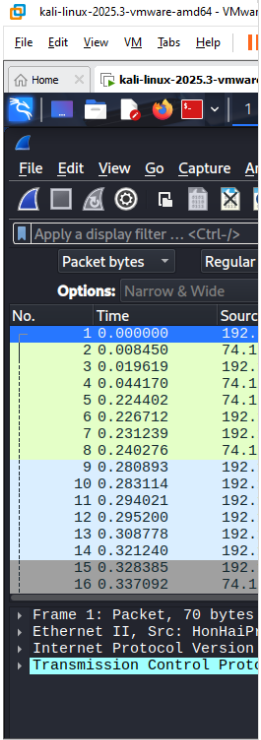
Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and the case title.
2. Executive conclusion. State the supported conclusion in a short, cautious paragraph. Do not name a suspect without explaining the evidence link.
3. Evidence acquisition and integrity. Record the source, file name, size, hash, working copy, and any published checksum comparison.
4. Method. Explain the sequence of Wireshark filters, packet inspection, and stream review you used.

5. Findings. Answer each of the eight investigation questions with citations to packet numbers, timestamps, screenshots, and evidence-log IDs.
6. Timeline. Show the relevant sequence of requests and responses using the timestamps from the PCAP.
7. Attribution assessment. Distinguish observed facts from inferences, give a confidence level, and explain the shared-network limitation and any other alternative explanation.
8. Appendices. Include labelled screenshots and the completed evidence log. Screenshots must show the active filter, packet list, relevant packet detail, and time where possible.

Evidence log template

Include a completed version of this table in your report or evidence appendix. You may add rows where needed.

Evidence ID	Packet or item	Finding	Why it matters	Screenshot or appendix
E01	94410			 The screenshot shows the Wireshark interface with a packet list on the left and packet details on the right. Packet 1 is selected, showing details for Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet list shows 16 packets, with packet 1 being the first packet in the capture.
E02				
E03				
E04				
E05				
E06				

Submission rules

- Work individually. Your interpretation, report wording, screenshots, and evidence log must be your own.
- Submit the report as SBTDF204_CaseStudy1_RegistrationNumber.pdf or .docx. If you submit separate screenshots, use SBTDF204_CaseStudy1_Evidence_RegistrationNumber.zip.

- Do not upload the original PCAP unless your lecturer specifically requests it. Submit only the report and the supporting evidence required for assessment.
- Treat the packet capture as controlled training material. Use it only for this authorised academic case study. Do not contact, identify, or make allegations about real people outside the scenario.
- Redact unrelated personal data that may appear in screenshots. Keep enough packet detail visible for the examiner to verify your finding.

Marking guide

The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score.

Criterion	What strong work demonstrates	Weight
Evidence acquisition and integrity	Records source, working copy, hash, and evidence handling clearly.	10%
Traffic discovery and filter use	Uses defensible filters and cites packets that identify the web service and client activity.	20%
Message correlation	Connects the HTTP evidence to the harassment content without overstating what the capture proves.	20%
Device and identity analysis	Uses MAC and cookie or browser evidence accurately and explains the link to the roster.	20%
Timeline and attribution assessment	Provides a chronological account, conclusion, confidence level, and relevant limitations.	20%
Report quality and evidence presentation	Writes clearly, labels screenshots, cites packet numbers, and submits a usable evidence log.	10%

Final check before submission

- Every major conclusion has a packet number, timestamp, filter, and screenshot or evidence-log reference.
- Your report explains why the client IP alone cannot establish the sender identity in a shared open wireless environment.
- Your timeline uses the time zone shown in Wireshark and identifies any conversion you made.
- Your conclusion stays within the scope of the evidence and clearly states the remaining uncertainty.